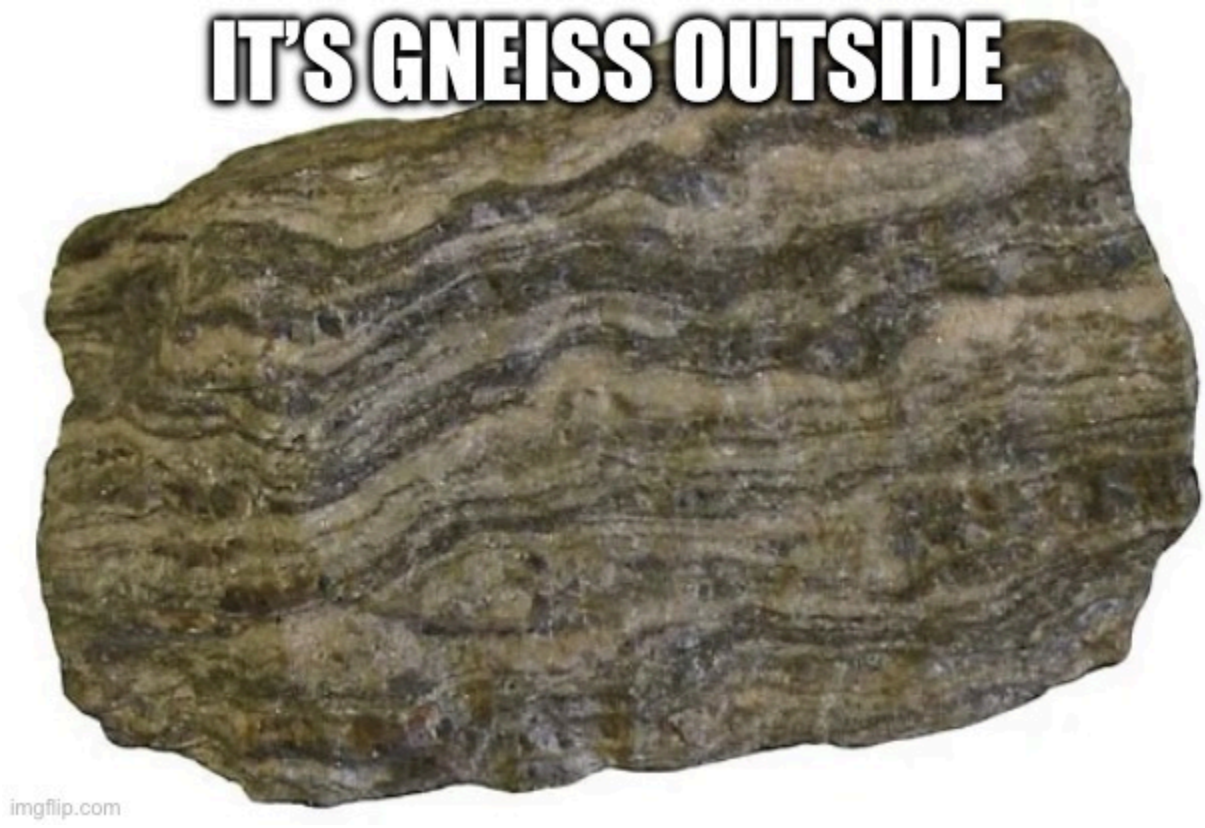
Rocks and Minerals

CSE 2025

Tesla STEM High School



Total: _____ / 167

Instructions: Each station is 3 minutes and 20 seconds long. Fill in the tables (each box is worth 1 point unless specified otherwise).

Station 1

Consider Bowen's Reaction Series.	
1) Which mineral crystallizes first?	
2) In the discontinuous series, which mineral crystallizes just before biotite?	
3) Which mineral is most resistant to chemical weathering?	
4) True or false: Minerals that are higher in the series are more stable at Earth's surface than those that crystallize lower in the series.	
5) Does calcium rich or sodium rich plagioclase have more silica?	
6) Based on the reaction series, which 3 minerals would you expect to find in gabbro? (1.5 points)	
7) Which branch does amphibole belong to?	
8) What mineral forms just before quartz?	
9) If a thick, basalt magma intrusion occurs, a layer of what mineral could you expect to find at the very bottom?	
10) Can a mafic magma evolve from a felsic one?	

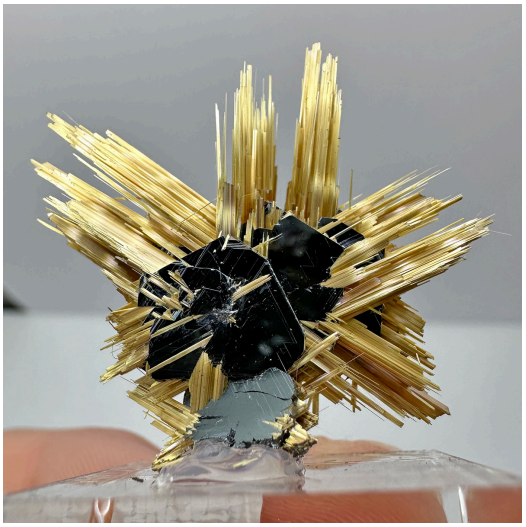
Station 2

11) Identify this mineral.	
12) What is its chemical formula?	
13) What color is its streak?	
14) Which rock is it most commonly found in?	
15) Who is it named after?	
16) In which era's rock beds is this mineral rare?	
17) What depositional environment is this formed in?	
18) Which element in water can turn calcite into this mineral?	
19) What is the process described in the previous question called?	
20) What type of bacteria living in what conditions can precipitate this mineral? (2 points)	

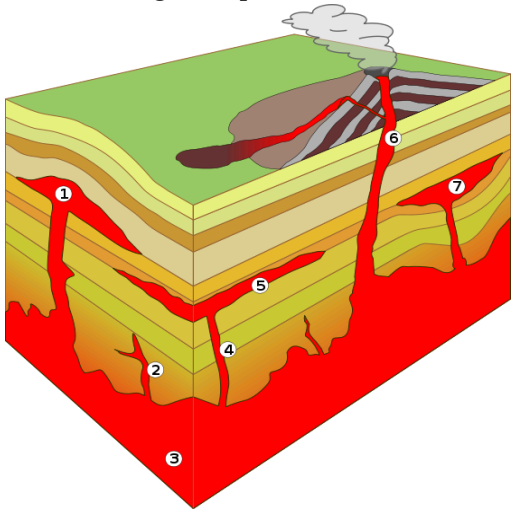
Station 3

21) Identify this mineral.	
22) What is its chemical formula?	
23) What color is its streak?	
24) What is its fracture?	
25) What is this mineral soluble in?	
26) What is its specific gravity?	
27) What type of ore contains large amounts of this mineral, a small amount of quartz, and a lot of black sand?	
28) In its natural mineral form, this mineral almost always has traces of (this mineral is also in the SciOly list):	
29) If added to food, can this mineral increase the nutritional content of food?	
30) What type of mining is used to extract this mineral from alluvial deposits?	

Station 4

	
31) Identify this mineral.	
32) What is its chemical formula?	
33) What is the color of its streak?	
34) Small needles of this mineral in gemstones are responsible for what optical phenomenon?	
35) This mineral is a source of what pigment? What color is this pigment? (2 points)	
36) What type of radiometric dating is this mineral commonly used for?	
37) Why can the mineral be used for the type of dating in the last question?	
38) This mineral is an accessory mineral. What does this mean?	
39) What are two polymorphs of this mineral?	
40) What elements, if present, can make this mineral appear black? (2 points)	

Station 5

Use this image for questions 41-44:  	
41) What is the igneous formation labeled by 1 in the diagram?	
42) What is the igneous formation labeled by 2 in the diagram?	
43) What is the igneous formation labeled by 5 in the diagram?	
44) What is the igneous formation labeled by 3 in the diagram?	
45) Is ice a mineral?	
46) Explain your answer above. (2 points)	
47) Which metamorphic facies has the lowest metamorphic grade?	
48) Which metamorphic facies occurs at low temperature and high pressure?	
49) What is the state mineral of Illinois?	
50) What is Nevada's state rock?	

Station 6

51) Identify this mineral.	
52) What is its chemical formula?	
53) What color is its streak?	
54) How does this mineral form?	
55) What variety of this mineral contains a high amount of arsenic impurity?	
56) If the arsenic substitution continues, what will the mineral become?	
57) What two elements are extracted from this mineral?	
58) Which country was this mineral first discovered in? What was it originally named?	
59) This mineral is commonly associated with the oxidation of what mineral?	
60) What country has the best specimen of this mineral?	

Station 7

61) Identify this mineral.	
62) What is its chemical formula?	
63) What color is its streak?	
64) What state(s) is this the state mineral of (state all states applicable): (1.5 points)	
65) What metal is this a primary ore for?	
66) What are the most common weathering products of this mineral?	
67) What is the variety of this mineral that contains silver? Where is it primarily found? (2 points)	
68) What is its crystal system?	
69) The pollution of which heavy metals are associated with mining of this mineral?	
70) A variety of this mineral containing Bismuth will have what cleavage?	

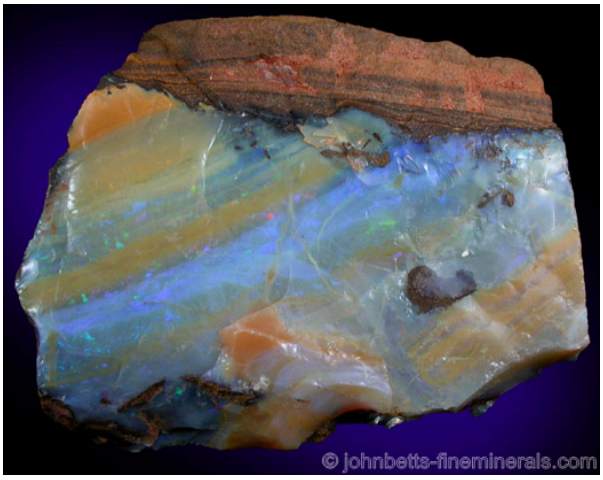
Station 8

71) Identify this mineral.	
72) What is its chemical formula?	
73) What is its crystal system?	
74) Is this mineral commonly found in sedimentary rocks? Why or why not? (2 points)	
75) This mineral is a nesosilicate. Describe the structure of nesosilicates.	
76) What are the two endmembers of the solution series that this mineral is in?	
77) Which endmember is more common?	
78) What happens to this mineral at a depth of 410 km?	
79) This mineral is the most abundant mineral in what part of the Earth?	
80) In the image, what is the rock that the mineral is embedded in?	

Station 9

81) Identify this mineral.	
82) What is its chemical formula?	
83) What is its crystal system?	
84) What type of plate boundary is this mineral typically found in?	
85) How did most of the deposits of this mineral in the US form?	
86) Why is this mineral so soft?	
87) What metamorphic rock is primarily composed of this mineral?	
88) What is this mineral's tenacity?	
89) Which form of this mineral is rarer than other forms?	
90) Is this mineral used as a conductor or an insulator?	

Station 10

	
91) Identify this mineraloid.	
92) What is its chemical formula?	
93) What color is its streak?	
94) What is this mineraloid composed of?	
95) Why can some specimen refract light while others can't?	
96) Can this mineraloid be scratched by a steel nail?	
97) What type of solution does this mineraloid form from?	
98) How does crazing affect some of these mineraloids?	
99) What are schillers?	
100) What is the range of water content in this mineraloid?	

Station 11

101) Identify this mineral.	
102) What is its chemical formula?	
103) What is its crystal system?	
104) What is it an ore for?	
105) Can this mineral be scratched by glass?	
106) What is metamictization?	
107) How is the color of this mineral changed by humans?	
108) What luster does this mineral have when it is radioactive?	
109) A darker brown or black color is due to what impurity?	
110) What is this mineral's tenacity?	

Station 12

111) Identify this rock.	
112) What is its texture?	
113) Is this rock intrusive or extrusive?	
114) Are mafic versions of this rock common?	
115) Lithium is extracted from specimens containing what mineral?	
116) How do bodies of this rock compare to bodies of other intrusive rocks?	
117) In a magma body, is this rock the first or last to crystallize?	
118) What rock's composition is most similar to this rock's composition?	
119) This rock forms from magmas rich in what?	
120) What is the role of the compound you identified in the previous question in promoting the growth of large crystals?	

Station 13

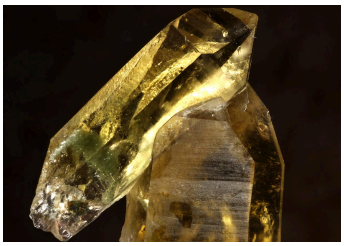
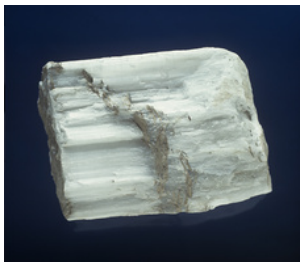
121) Identify this rock.	
122) What is its texture?	
123) In what depositional environment are the grains well rounded and poorly sorted?	
124) In what depositional environment are the grains well rounded and poorly sorted?	
125) What is the matrix?	
126) What does the rounding of clasts indicate about their history?	
127) When the grains are poorly sorted, what does this indicate about the past environment?	
128) What does it mean when this rock is clast supported?	
129) What does it mean when this rock is matrix supported?	
130) Which rock is similar to this rock but has angular fragments?	

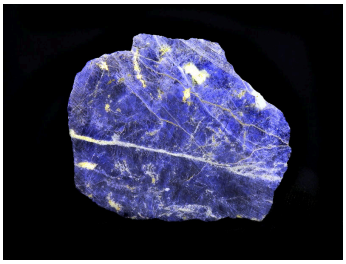
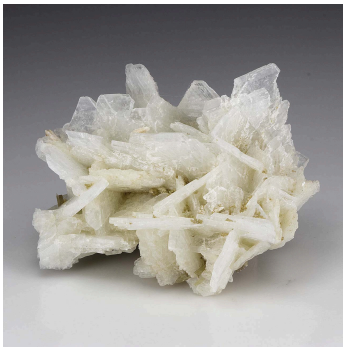
Station 14

131) Identify this rock.	
132) What is its texture?	
133) What is its parent rock?	
134) What type of metamorphism usually forms this rock?	
135) What temperatures does this rock usually form under?	
136) What pressures does this rock usually form under?	
137) Which minerals is this rock primarily composed of?	
138) In the image, which mineral makes up the black portions?	
139) Which plate boundary is this rock associated with?	
140) What are high quality specimen used for?	

Station 15

Identify each rock/mineral.

			
141)	142)	143)	144)
145)	146)	147)	148)
149)	150)	151)	152)

			
153)	154)	155)	156)
			
157)	158)	159)	160)

Captain’s Summer Exchange - Tesla STEM High School - Repaired practice edition

Division C | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

27. (6 points) Authored replacement for source item Question 27: Solve this case and explain your reasoning: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

Answer: _____
_____
_____

60. (6 points) Authored replacement for source item Question 60: Solve this case and explain your reasoning: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

Answer: _____
_____
_____

70. (4 points) Authored replacement for source item Question 70: Differentiate mineral and rock cleavage.

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

Answer: _____
_____
_____

95. (6 points) Authored replacement for source item Question 95: Solve this case and explain your reasoning: A massive specimen lacks visible faces. Does it lack crystalline order?

Answer: _____
_____
_____

96. (6 points) Authored replacement for source item Question 96: Solve this case and explain your reasoning: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

Answer: _____
_____
_____

123. (4 points) Authored replacement for source item Question 123: What is a pseudomorph?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

Answer: _____
_____
_____

124. (4 points) Authored replacement for source item Question 124: What do vesicles record?

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

Answer: _____
_____
_____